

Suspended Films Research

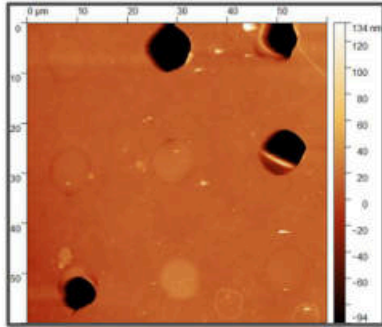
By Anderson and Jun

Observations

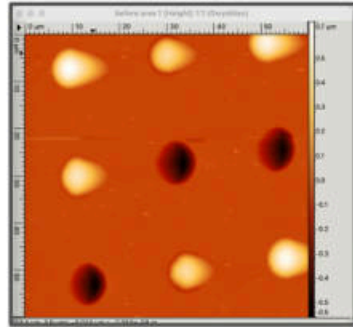
1. June 29th - July 10th, we observed that previously bulged/flat suspended films over wells deflected down after 5 mins of vacuuming, 2 minutes deflected some wells, 5 minutes definitively caused deflection downwards, 10 minutes and 15 minutes were tested and however likely caused the film to stick to the inside of the wells, which caused the film to slowly bulge back up over time.
2. From March to July, we also found that certain films degraded and bulged down due to air leakage from loss of adhesion.

Examples of Film Deflection

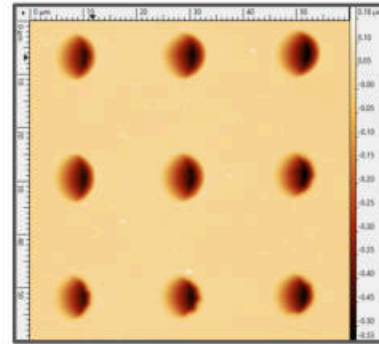
Flat Wells



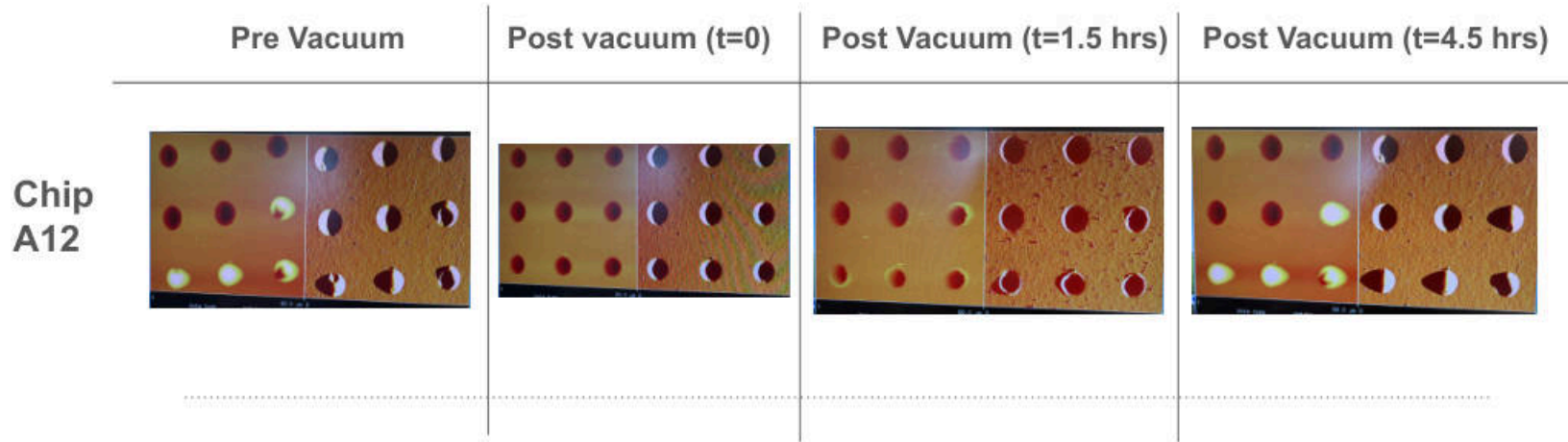
Bulged Up Wells



Bulged Down Wells



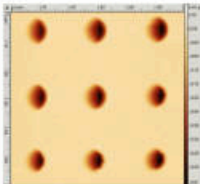
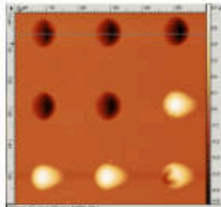
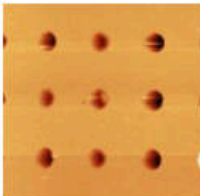
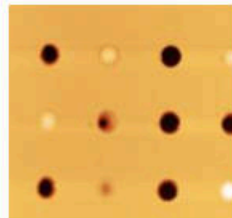
Before/After Vacuum (Bulged Up Wells)

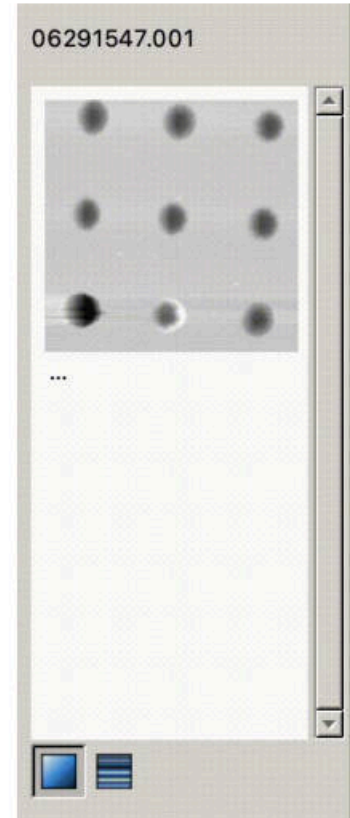


Observations:

After vacuuming, the bulged up film seems to deflate, then bulge back up after resting.

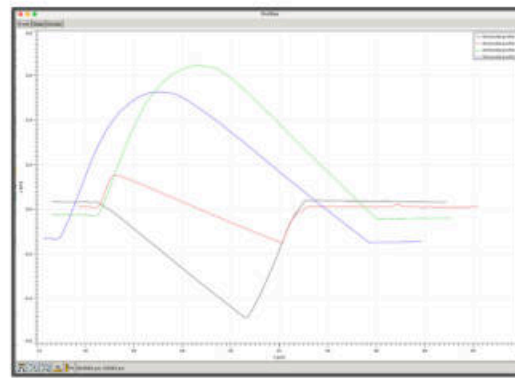
Rate of Recovery After Vacuum on Film

	Post Vacuum	Post Resting	Start	End																		
Chip A12			<table><tr><td>-441.5 nm</td><td>-496.1 nm</td><td>-510.0 nm</td></tr><tr><td>-483.1 nm</td><td>-498.7 nm</td><td>-496.2 nm</td></tr><tr><td>-480.3 nm</td><td>-511.4 nm</td><td>-525.7 nm</td></tr></table>	-441.5 nm	-496.1 nm	-510.0 nm	-483.1 nm	-498.7 nm	-496.2 nm	-480.3 nm	-511.4 nm	-525.7 nm	<table><tr><td>-516.1 nm</td><td>-542.6 nm</td><td>-559.8 nm</td></tr><tr><td>-517.6 nm</td><td>-552.0 nm</td><td>653 nm</td></tr><tr><td>660 nm</td><td>671 nm</td><td>535 nm</td></tr></table>	-516.1 nm	-542.6 nm	-559.8 nm	-517.6 nm	-552.0 nm	653 nm	660 nm	671 nm	535 nm
-441.5 nm	-496.1 nm	-510.0 nm																				
-483.1 nm	-498.7 nm	-496.2 nm																				
-480.3 nm	-511.4 nm	-525.7 nm																				
-516.1 nm	-542.6 nm	-559.8 nm																				
-517.6 nm	-552.0 nm	653 nm																				
660 nm	671 nm	535 nm																				
Chip F1			<table><tr><td>-218.0 nm</td><td>-162.9 nm</td><td>-273.4 nm</td></tr><tr><td>-147.0 nm</td><td>-135.7 nm</td><td>-230.9 nm</td></tr><tr><td>-157.4 nm</td><td>-150.6 nm</td><td>-192.9 nm</td></tr></table>	-218.0 nm	-162.9 nm	-273.4 nm	-147.0 nm	-135.7 nm	-230.9 nm	-157.4 nm	-150.6 nm	-192.9 nm	<table><tr><td>-432.8 nm</td><td>116.5 nm</td><td>-429.3 nm</td></tr><tr><td>106.9 nm</td><td>-433.8 nm</td><td>-421.1 nm</td></tr><tr><td>-438.0 nm</td><td>-132.0 nm</td><td>-428.0 nm</td></tr></table>	-432.8 nm	116.5 nm	-429.3 nm	106.9 nm	-433.8 nm	-421.1 nm	-438.0 nm	-132.0 nm	-428.0 nm
-218.0 nm	-162.9 nm	-273.4 nm																				
-147.0 nm	-135.7 nm	-230.9 nm																				
-157.4 nm	-150.6 nm	-192.9 nm																				
-432.8 nm	116.5 nm	-429.3 nm																				
106.9 nm	-433.8 nm	-421.1 nm																				
-438.0 nm	-132.0 nm	-428.0 nm																				

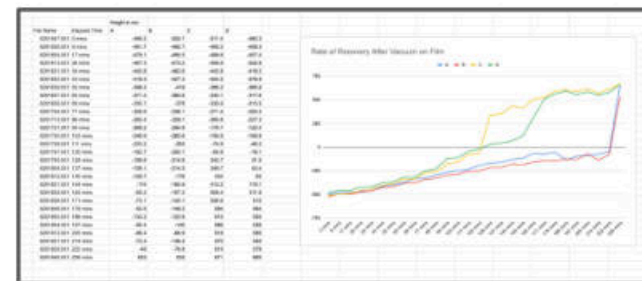
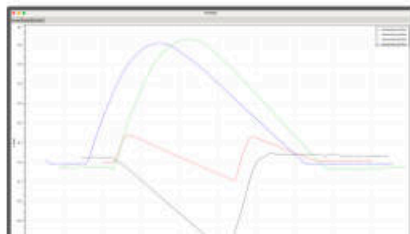


[Rate of Recovery After Vacuum on Film](#) Google Sheets

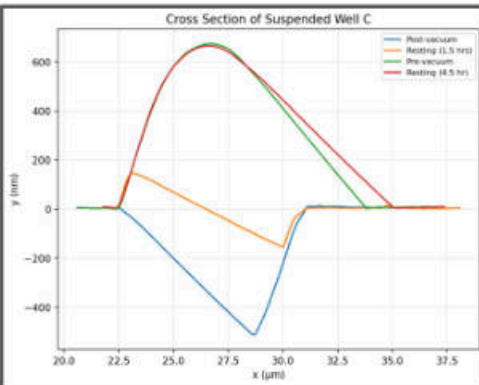
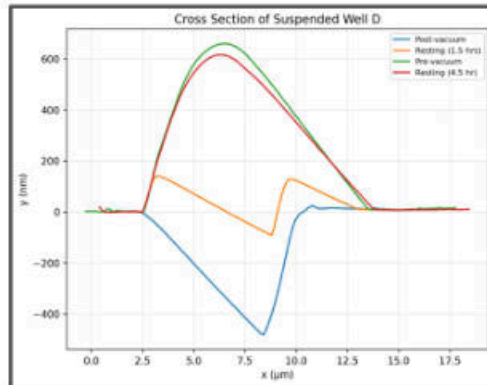
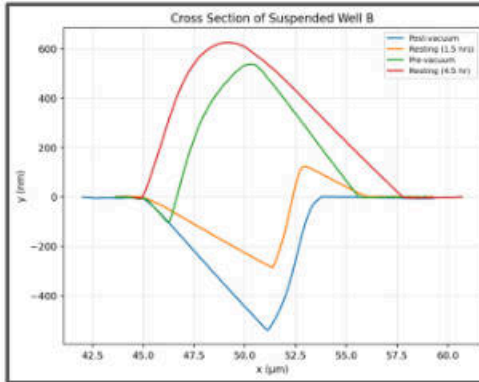
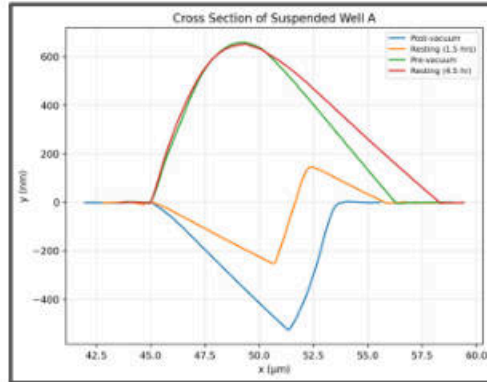
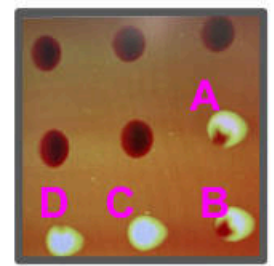
Suspended Film C



Suspended Film D



Offset Cross Section Graph of Inflated Wells

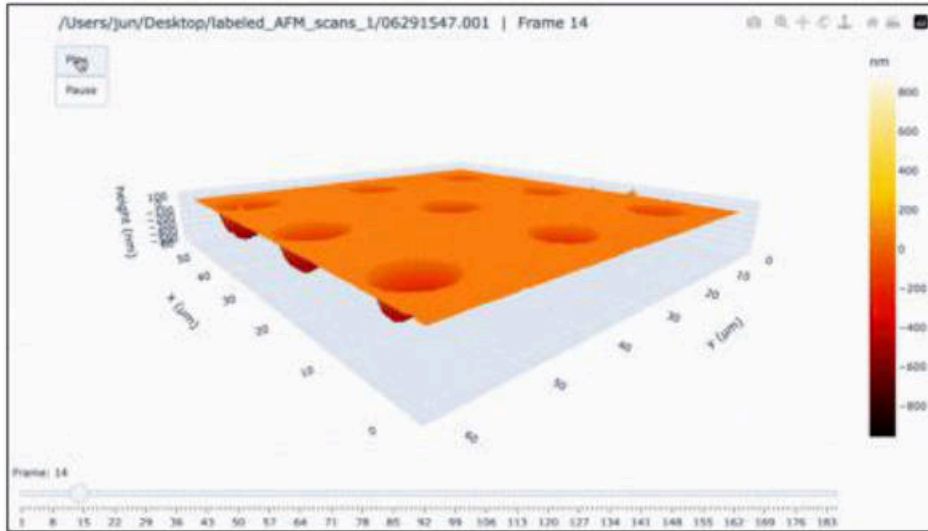


Observations:
For some wells, the film stuck to the inside may have delaminated while inflating causing it to increase the max deflection.

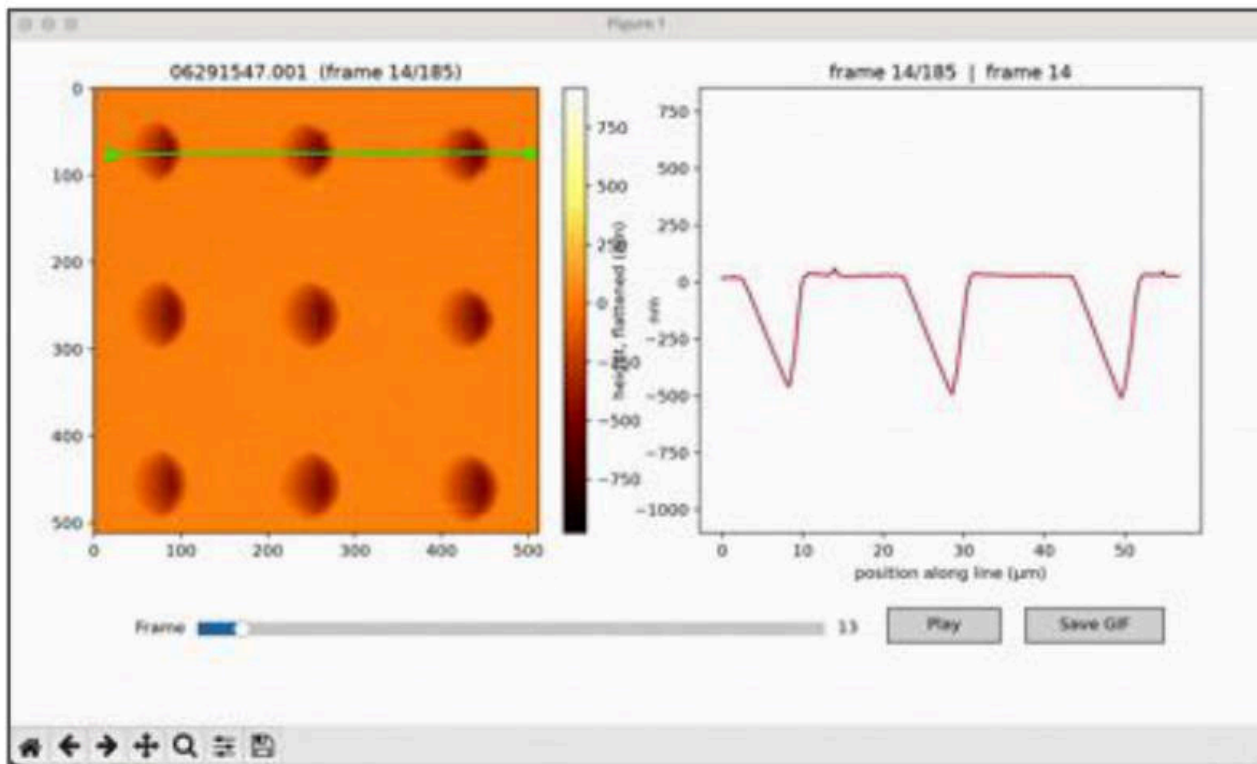
3D Visualization Using Python matplotlib and Gwyddion

Chip A12

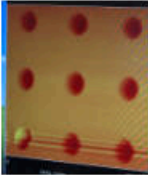
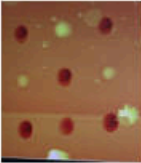
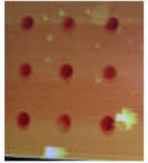
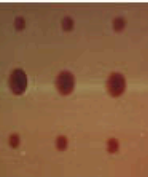
Chip F1



Cross Section Visualization using Python matplotlib



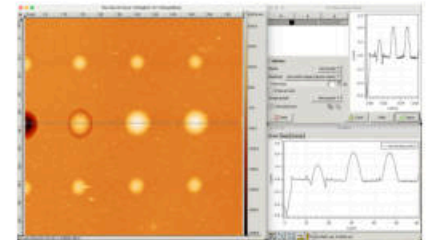
Before/After Vacuum of 10 minutes

	Pictures		Heights (nm)					
	Before	After	Before			After		
Chip A1			80.5 nm	124.5 nm	201.5 nm	-290.5 nm	-299.9 nm	-297.3 nm
			38.2 nm	95.0 nm	179.8 nm	-296.6 nm	-296.0 nm	-289.9 nm
			-173.5 nm	-142.5 nm	108.5 nm	-295.0 nm	-280.5 nm	-252.1 nm
Chip C1			-352.1 nm	176.5 nm	-351.7 nm	-362.1 nm	-202.3 nm	-351.0 nm
			89.9 nm	-362.5 nm	100.7 nm	-201.1 nm	-362.5 nm	-211.4 nm
			-400.7 nm	-166.9 nm	-367.6 nm	-378.0 nm	-152.0 nm	-334.0 nm
Chip D1			147.9 nm	137.9 nm	125.2 nm	-214.6 nm	-227.3 nm	-201.0 nm
			125.0 nm	218.7 nm	222.4 nm	-446.6 nm	-341.1 nm	-340.3 nm
			142.1 nm	131.8 nm	125.0 nm	-211.0 nm	-212.4 nm	-206.9 nm

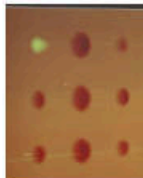
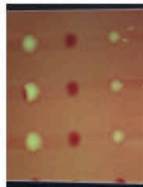
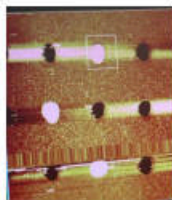
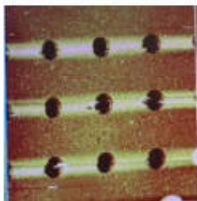
Observations:

- Bulged up films deflated.
- Bulged down films either remained same or deflected further.

Example of analysis of D1



Before/After Vacuum of 10 minutes

	Pictures		Heights (nm)																			
	Before	After	Before	After																		
Chip E3			<table><tr><td>188.7 nm</td><td>-225.2 nm</td><td>153.9 nm</td></tr><tr><td>-160.6 nm</td><td>153.0 nm</td><td>-382.9 nm</td></tr><tr><td>-159.9 nm</td><td>158.6 nm</td><td>-147.6 nm</td></tr></table>	188.7 nm	-225.2 nm	153.9 nm	-160.6 nm	153.0 nm	-382.9 nm	-159.9 nm	158.6 nm	-147.6 nm	<table><tr><td>186.7 nm</td><td>-269.9 nm</td><td>-144.4 nm</td></tr><tr><td>-153.5 nm</td><td>-230.8 nm</td><td>-186.8 nm</td></tr><tr><td>-143.3 nm</td><td>-223.3 nm</td><td>-157.1 nm</td></tr></table>	186.7 nm	-269.9 nm	-144.4 nm	-153.5 nm	-230.8 nm	-186.8 nm	-143.3 nm	-223.3 nm	-157.1 nm
	188.7 nm	-225.2 nm	153.9 nm																			
-160.6 nm	153.0 nm	-382.9 nm																				
-159.9 nm	158.6 nm	-147.6 nm																				
186.7 nm	-269.9 nm	-144.4 nm																				
-153.5 nm	-230.8 nm	-186.8 nm																				
-143.3 nm	-223.3 nm	-157.1 nm																				
			<table><tr><td>165.7 nm</td><td>-153.0 nm</td><td>108.1 nm</td></tr><tr><td>171.4 nm</td><td>-150.7 nm</td><td>87.0 nm</td></tr><tr><td>172.6 nm</td><td>-143.5 nm</td><td>112.1 nm</td></tr></table>	165.7 nm	-153.0 nm	108.1 nm	171.4 nm	-150.7 nm	87.0 nm	172.6 nm	-143.5 nm	112.1 nm	<table><tr><td>-426.8 nm</td><td>-211.6 nm</td><td>-211.0 nm</td></tr><tr><td>212.4 nm</td><td>-211.6 nm</td><td>-209.7 nm</td></tr><tr><td>-433.3 nm</td><td>-212.4 nm</td><td>-206.9 nm</td></tr></table>	-426.8 nm	-211.6 nm	-211.0 nm	212.4 nm	-211.6 nm	-209.7 nm	-433.3 nm	-212.4 nm	-206.9 nm
165.7 nm	-153.0 nm	108.1 nm																				
171.4 nm	-150.7 nm	87.0 nm																				
172.6 nm	-143.5 nm	112.1 nm																				
-426.8 nm	-211.6 nm	-211.0 nm																				
212.4 nm	-211.6 nm	-209.7 nm																				
-433.3 nm	-212.4 nm	-206.9 nm																				
Chip F1			<table><tr><td>-262.0 nm</td><td>79.4 nm</td><td>-233.5 nm</td></tr><tr><td>112.7 nm</td><td>-127.8 nm</td><td>-233.0 nm</td></tr><tr><td>-240.2 nm</td><td>83.5 nm</td><td>-312.0 nm</td></tr></table>	-262.0 nm	79.4 nm	-233.5 nm	112.7 nm	-127.8 nm	-233.0 nm	-240.2 nm	83.5 nm	-312.0 nm	<table><tr><td>-179.3 nm</td><td>-137.7 nm</td><td>-197.9 nm</td></tr><tr><td>-110.4 nm</td><td>-88.5 nm</td><td>-198.0 nm</td></tr><tr><td>-140.5 nm</td><td>-131.7 nm</td><td>-142.3 nm</td></tr></table>	-179.3 nm	-137.7 nm	-197.9 nm	-110.4 nm	-88.5 nm	-198.0 nm	-140.5 nm	-131.7 nm	-142.3 nm
	-262.0 nm	79.4 nm	-233.5 nm																			
112.7 nm	-127.8 nm	-233.0 nm																				
-240.2 nm	83.5 nm	-312.0 nm																				
-179.3 nm	-137.7 nm	-197.9 nm																				
-110.4 nm	-88.5 nm	-198.0 nm																				
-140.5 nm	-131.7 nm	-142.3 nm																				

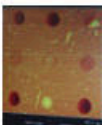
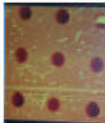
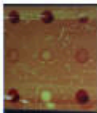
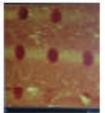
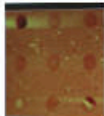
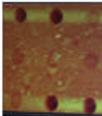
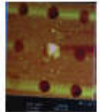
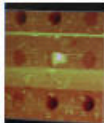
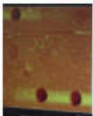
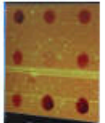
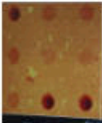
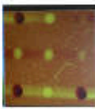
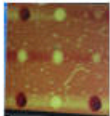
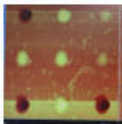
Observations:

- Bulged up films deflated.
- Bulged down films either remained same or deflected further down.

Before/After Vacuum of 5 minutes

Pictures

Heights (nm)

Chip G1	Before	After Vacuum	After resting
Area 1			
Area 2			
Area 3			
Area 4			
Area 5			

Area 1 - 07121457.001 & 07121553.001 & 07131425.001

Before				After				After Resting 22 hrs			
		-50				-225					21
-26	18	34	-283	-278	-286	30	62	18			
	120				-225		153				

Area 3 - 07121520.001 & 07121618.001 & 07131502.001

Before				After				After Resting 22 hrs			
		-27				-207					-55
-34	-38	-31	-212	514	-212	-45	302	-35			

Area 2 - 07121507.001 & 07121610.001 & 07131453.001

Before				After				After Resting 22 hrs			
		-37	-40			45					-40
-39	-35	-47	-165	-178	-137	-45	-36	-47			
50	-28	-42	-134	-86	140	59	-32	42			

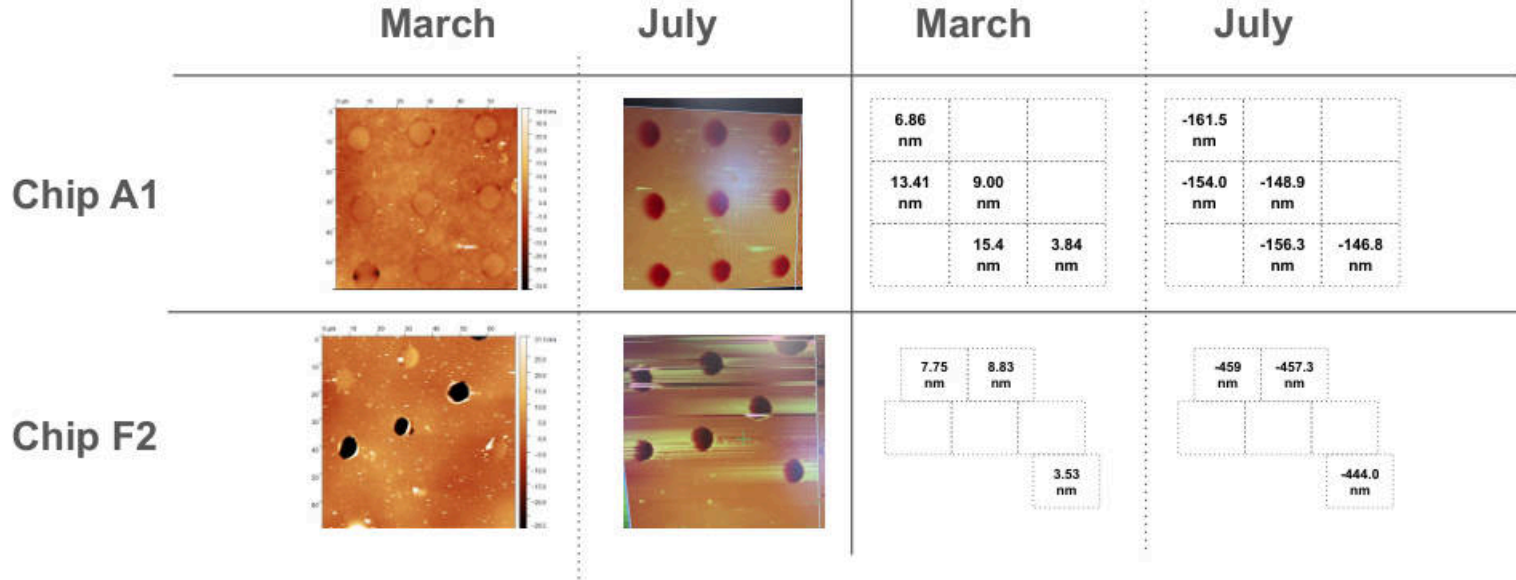
Area 4 - 07121529.001 & 07121627.001 & 07131510.001

Before				After				After Resting 22 hrs			
		-23	-33			-144	-159			-31	-38
-32	-22	9	-147	22	-147	-44	-24	-24			
24											

Area 5 - 07121536.001 & 07121634.001 & 07131523.001

Before				After				After Resting 22 hrs			
		150	34			186	112			173.9	48.6
144	135	56	177	171	88	158.74	162.3	79.9			

Films that degraded from March/April to July



Observations: Films degraded due to unknown reasons, not touched or experimented on since March/April.

Before/After Pressurizing

Pictures and Heights Below

Chip A1

Before

After Vacuum

After Pressure

After resting 24 hrs

80.5 nm	124.5 nm	201.5 nm
38.2 nm	95.0 nm	179.8 nm
-173.5 nm	-142.5 nm	100.5 nm

-290.5 nm	-299.9 nm	-297.3 nm
-296.6 nm	-298.0 nm	-289.9 nm
-295.0 nm	-280.5 nm	-252.1 nm

186.9 nm	247.31 nm	260.4 nm
190.4 nm	207.3 nm	239.3 nm
113.6 nm	176.6 nm	197.0 nm

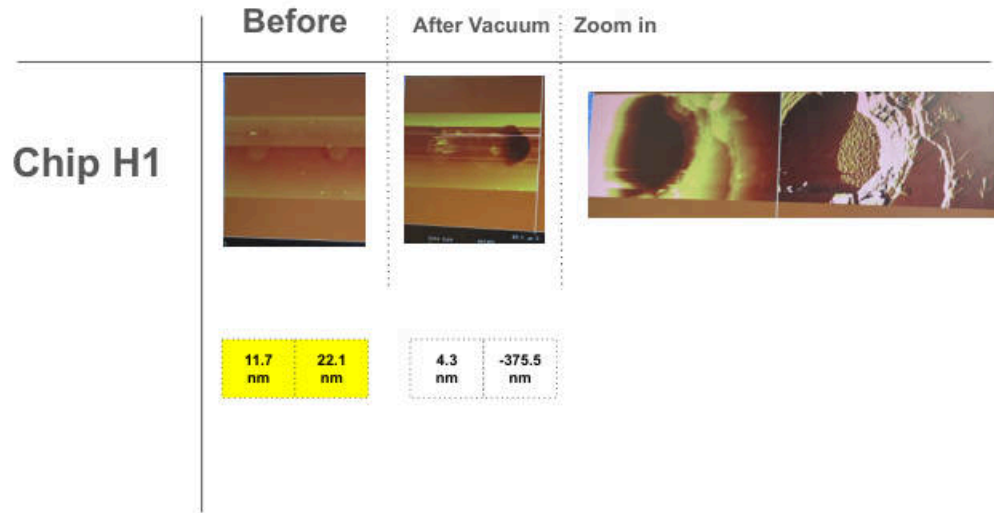
122.9 nm	74.5 nm	-202.8 nm
191.1 nm	124.1 nm	79.7 nm
211.8 nm	196.1 nm	96.3 nm

Observations:

After pressurizing the already vacuumed chip for roughly 24 hours in Ar gas, it deflected higher than it was before. When resting however, the deflection reduced gradually.

Before/After Vacuum of 20 minutes

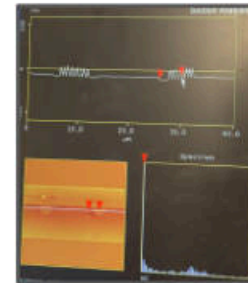
Pictures and Heights Below



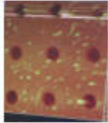
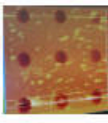
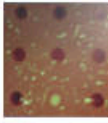
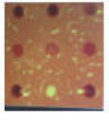
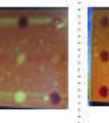
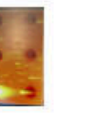
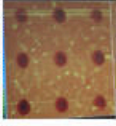
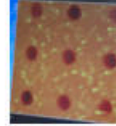
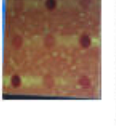
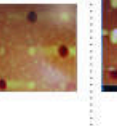
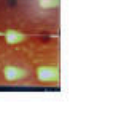
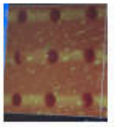
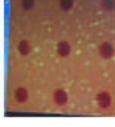
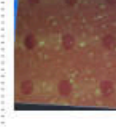
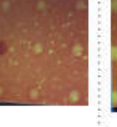
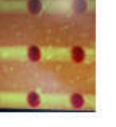
Observations:

Films deflections were spiky and were read from the lowest point of the spikes as these were likely caused by the film vibrating due to the tip.

The film likely tore off and was damaged due to longer vacuuming, both films over each well likely has holes too.



Before/After Ozone Cleaning AFM Scans

		After Ozone					
Chip G1	After vacuuming	30 secs	2 mins	3 mins	4 mins	5 mins	10 mins
Area 1							
Area 2							
Area 3							

Observations:

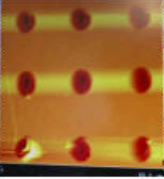
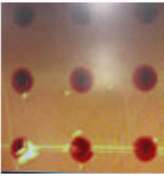
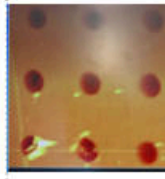
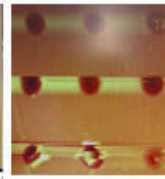
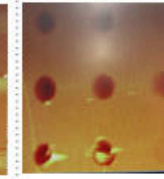
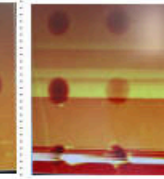
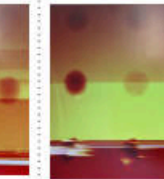
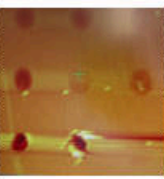
Ozone likely shaved off layers causing the suspended films to allow more air to permeate, until too much was shaved off, which caused them to burst at 10 mins.

Before/After Ozone Cleaning Height Data

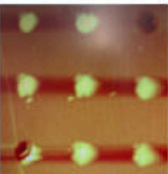
Heights (nm)

(In nm)																								
G1 area 1 - 07181412.001 & 07181530.001 & 07181602.001 & 07181653.001 & 07181721.001 & 07181755.001 & 07181924.001 & 17182002.001																								
					Vacuum					After Ozone Cleaning														
Before					After Vacuum					After 30 sec					After 2 mins					After 3 mins				
-12	39	-50	-281	-265	-193	-283	-240	-175	-190	-102	-188	-59	-20	-94	57	88	26	74	103	55	-293	-196	-89	
	184			-248			-33			98			111			203			199				276	
G1 area 2 - 07181423.001 & 07181537.001 & 07181623.001 & 07181701.001 & 07181728.001 & 07181804.001 & 17181931.001 & 07182016.001																								
					Vacuum					After Ozone Cleaning														
Before					After Vacuum					After 30 sec					After 2 mins					After 3 mins				
-46		-44	-230		-145	-232	-241	-154		-155	-38		-53	-44		22	-31		70	312		364		
-62	-53		-300	-254		-302	-278	-250	-222		-57	-56		31	13		78	61		338	373			
	-45	-55		-250	-165		-305	-239		-210	-142		-83	-92		19	15		68	66		367	368	
G1 area 3 - 07181432.001 & 07181547.001 & 07181632.001 & 07181709.001 & 07181735.001 & 07181821.001 & 07181938.001 & 07182024.001																								
					Vacuum					After Ozone Cleaning														
Before					After Vacuum					After 30 sec					After 2 mins					After 3 mins				
-47	-47	-54	-289	-270	-218	-291	-268	-206	-253	-252	-231	-59	-52	-64	-33	-36	-36	69	70	61	-456	-418	-347	
-51	-41	-47	-222	-260	-189	-225	-246	-181	-226	-246	-223	-189	-57	-62	-166	-34	-38	-145	63	59	-468	-403	-340	
-50	-47	-52	-295	-264	-175	-294	-248	-172	-265	-243	-238	-62	-62	-69	-31	-34	-35	54	48	62	-475	-419	-342	

Before/After Ozone Cleaning AFM Images

		After Ozone						
Chip A1	Before Vacuuming	After vacuuming	1 min	2 mins	3 mins	4 mins	5 mins	6 mins
	Area 1							
	7 mins	8 mins	9 mins	10 mins	11 mins	12 mins		
								

Before/After Ozone Cleaning Optical Microscope

		After Ozone						
Chip A1	Before Vacuuming	After vacuuming	1 min	2 mins	3 mins	4 mins	5 mins	6 mins
Area 1								
	7 mins	8 mins	9 mins	10 mins	11 mins	12 mins		
								

Before/After Ozone Cleaning Height Data

(in nm)

A1 - 07191537.001 & 07191611.001 & 07191631.001 & 07191649.001 & 07191710.001 & 07191743.001 & 07191906.001 & 07191922.001 & 07191959.001 & 07192014.001

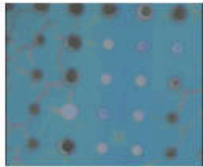
			Vacuum			After Ozone Cleaning																	
Before Vacuuming			After Vacuuming			1 min			2 mins			3 mins			4 mins			5 mins			6 mins		
271	277	-195	-377	-356	-284	-389	-376	-332	-330	-281	-188	-315	-293	-165	-223	-171	-34	-167	-139	-38	-131	-96	-41
305	293	252	-429	-328	-251	-459	-359	-335	-342	-261	-203	-434	-280	-304	-286	-170	-108	-347	-80	-125	-309	-47	-106
	296	274		-280	-212		-359	-293		-137	-182		-206	-140		-125	10		-164	10		-246	10

[illegible]

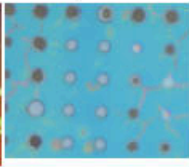
Comparison of Plasma/Ozone/Vacuum

Plasma

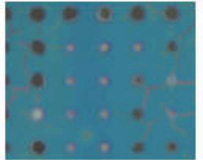
Before



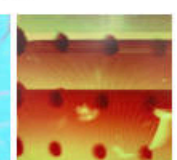
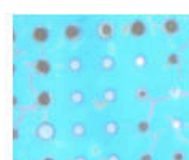
After (2 seconds)



After (1 second)

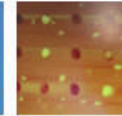
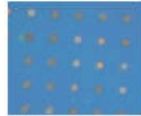


After (3 seconds)

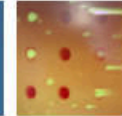
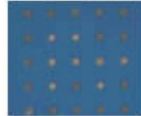


Vacuum

Before



After vacuum 5 mins

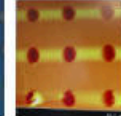


After vacuum 10 mins

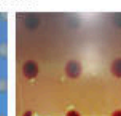


Ozone

Before



After ozone 1 min



After ozone 8 mins



Observations:

- By doing 2 conditions, one with plasma and without plasma but still vacuuming, the damage of the wells is because of the plasma treatment, and not due to vacuuming.
- For the ozone treatment, we see that the film over the substrate (non-well areas, areas with no wells) are not being treated the same way the bulges are, i.e no color difference on flat parts of film vs bulged parts of film.
- Gemini suggested that for ozone treatment, this occurs because the substrate (Si) acts as a heat sink and cools the film, whereas the bulged parts are able to heat freely.

Oxidation Experiments

By Andreson and Jun

link to journal: https://docs.google.com/presentation/d/1Cnz8xTmDUB193G_ce4yGOaHQ2YwT-uCv4bEkWoIVpyA/edit?slide=id.g3f61325622a_1_3#slide=id.g3f61325622a_1_3

Three experiments

1. Initial test experiment: showed less oxidation from the
2. Overnight experiment: the copper foil had both polystyrene layer as well as the 2DPA-1 layer, but after learning that the polystyrene has a crystallization point of 100C, we chose to run one that heated the foil for multiple days at around 72C.
3. Delaminated experiment: we removed the polystyrene layer to test just the effects of 2DPA-1 on oxidation.

1. Initial experiment

The experiment was run at 125C for 3 hours and showed very little oxidation both bare copper and under the 2DPA-1.

Before



After



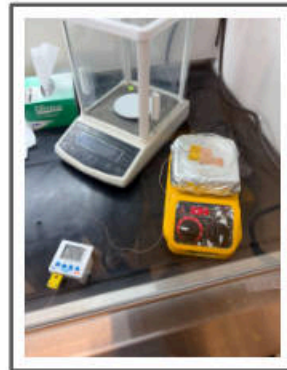
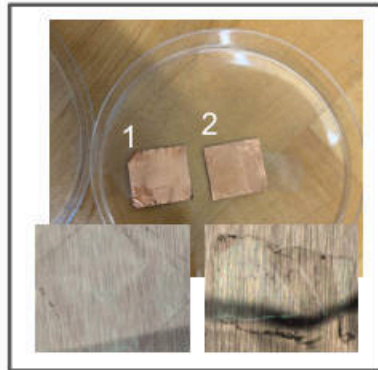
2. Overnight Experiment

One control and two film-coated copper foils were heated at 73°C for several days. This specific temperature was chosen to prevent burning the colored polystyrene layer, which sits above the opaque, dark-outlined 2DPA-1 layer. To isolate the effects and better confirm the results we took off the polystyrene using chloroform and hexane.

10 nm film: $\lambda = 4 \times 1.59 \times 10 = 63.6$
nm (Ultraviolet, not visible)

35 nm film: $\lambda = 4 \times 1.59 \times 35 = 222.6$
nm (Ultraviolet, not visible)

1000 nm (1 μm) film: $\lambda = \frac{2 \times 1.59 \times 1000}{m} =$
636 nm (Red at $m = 5$), 530 nm (Green
at $m = 6$), and 454 nm (Blue at $m = 7$)



3. Delaminated experiment

After removing the layer, the 2DPA-1 layer seemed to prevent some oxidation but on a select area, notice the square border in the after picture.

Before



After



Wavelength of Red Light (λ): ~ 650 nm
Refractive Index of Copper Oxide (Cu_2O) (n): ~ 2.7

Using the simple thin-film interference formula for the first visible color ($t = \frac{\lambda}{4n}$):

- $t = \frac{650}{4 \times 2.7}$
- $t = \frac{650}{10.8}$
- $t \approx 60$ nm

When the copper foil turns red, the oxide layer is roughly 60 nanometers thick.

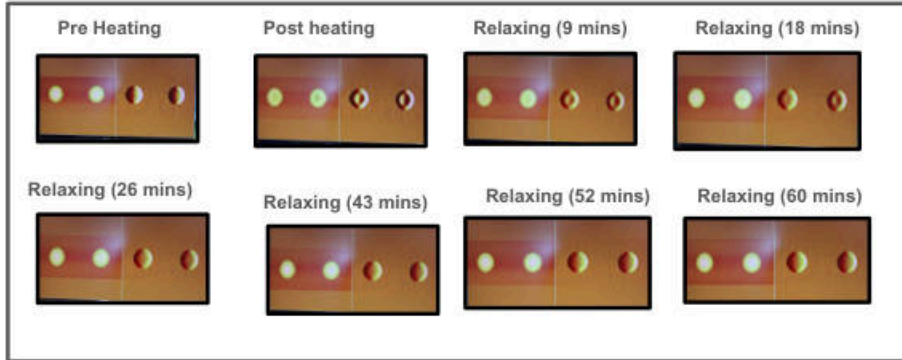
Temperature on Suspended Films

By Aviva and Jun

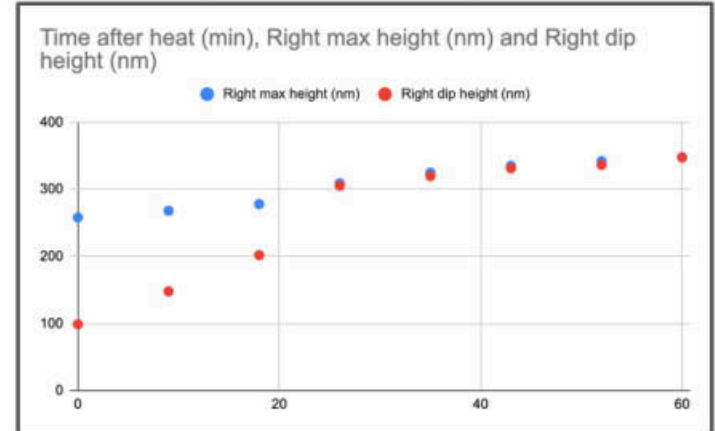
link to data: <https://docs.google.com/spreadsheets/d/1ljafiDhDFx5-vXzwxVCLiUeCL5Ryj9HTtEAYjzdHuuO/edit?gid=0#gid=0>

Experiment 1: Buckling after heating

We heated the chip for 10 mins at 65C and found that, immediately after heating, we scanned the films and found a dip/buckling effect, and as we cool it, it relaxes back into its shape.



A	B	C
Time after heat (min)	Right max height (nm)	Right dip height (nm)
0	258	99
9	268	148
18	278	202
26	309	305.47
35	325	319.75
43	335	331.35
52	342	336.35
60	348	347.38

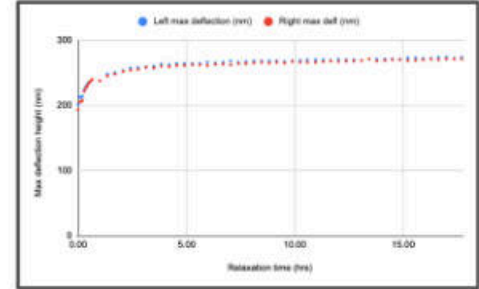
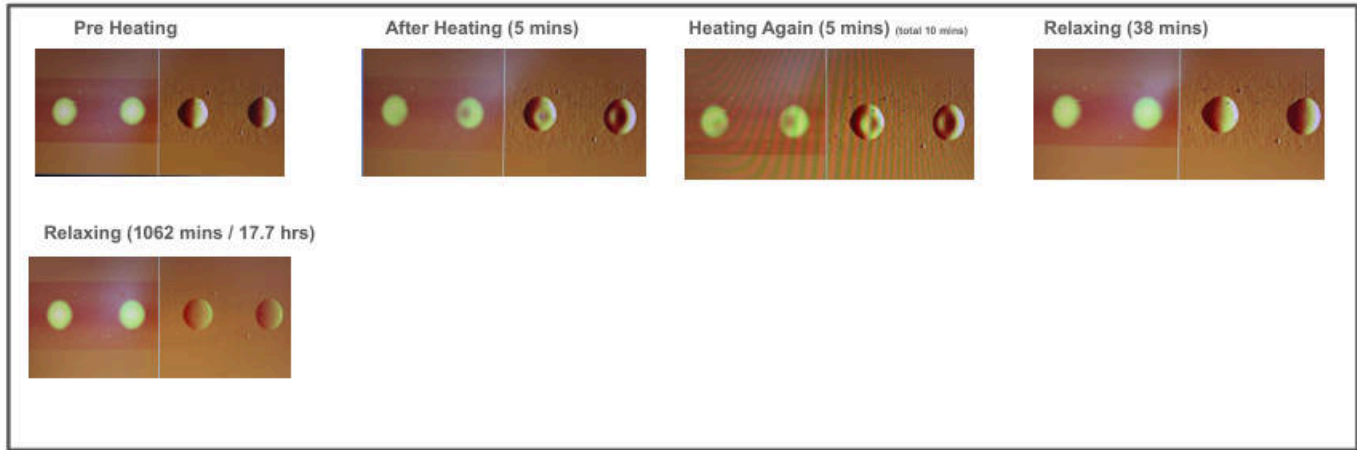


Experiment 2: Buckling due to Interlayer Creep

On 4/22, we heated the same sample at 60C for 5 minutes, twice, measuring both before and after, then scanned the relaxation overnight for a longer period. Same observations as before, buckling effect, then relaxes into original shape. As you rest the film, the buckled part increases and returns to its original height.

Gemini suggested “the pressure inside the cavity is actually pushing upward harder” and that interlayer creep is an explanation for the buckling effect. As heat is applied to the film, the H bonds loosen the sheets of stacked 2DPA-1 allowing the shape to change.

Heating phase (60°C) (minutes)					
Time (mins)	Left max deflection (mm)	Left max defl (mm)	Right max defl (mm)	Right max defl (mm)	Right max defl (mm)
0	0.00	0.00	0.00	0.00	0.00
5	0.10	0.10	0.10	0.10	0.10
10	0.20	0.20	0.20	0.20	0.20
Relaxing phase (60°C) (minutes)					
Time (mins)	Left max deflection (mm)	Left max defl (mm)	Right max defl (mm)	Right max defl (mm)	Right max defl (mm)
0	0.00	0.00	0.00	0.00	0.00
5	0.10	0.10	0.10	0.10	0.10
10	0.20	0.20	0.20	0.20	0.20
15	0.30	0.30	0.30	0.30	0.30
20	0.40	0.40	0.40	0.40	0.40
25	0.50	0.50	0.50	0.50	0.50
30	0.60	0.60	0.60	0.60	0.60
35	0.70	0.70	0.70	0.70	0.70
40	0.80	0.80	0.80	0.80	0.80
45	0.90	0.90	0.90	0.90	0.90
50	1.00	1.00	1.00	1.00	1.00
55	1.10	1.10	1.10	1.10	1.10
60	1.20	1.20	1.20	1.20	1.20
65	1.30	1.30	1.30	1.30	1.30
70	1.40	1.40	1.40	1.40	1.40
75	1.50	1.50	1.50	1.50	1.50
80	1.60	1.60	1.60	1.60	1.60
85	1.70	1.70	1.70	1.70	1.70
90	1.80	1.80	1.80	1.80	1.80
95	1.90	1.90	1.90	1.90	1.90
100	2.00	2.00	2.00	2.00	2.00
105	2.10	2.10	2.10	2.10	2.10
110	2.20	2.20	2.20	2.20	2.20
115	2.30	2.30	2.30	2.30	2.30
120	2.40	2.40	2.40	2.40	2.40
125	2.50	2.50	2.50	2.50	2.50
130	2.60	2.60	2.60	2.60	2.60
135	2.70	2.70	2.70	2.70	2.70
140	2.80	2.80	2.80	2.80	2.80
145	2.90	2.90	2.90	2.90	2.90
150	3.00	3.00	3.00	3.00	3.00
155	3.10	3.10	3.10	3.10	3.10
160	3.20	3.20	3.20	3.20	3.20
165	3.30	3.30	3.30	3.30	3.30
170	3.40	3.40	3.40	3.40	3.40
175	3.50	3.50	3.50	3.50	3.50
180	3.60	3.60	3.60	3.60	3.60
185	3.70	3.70	3.70	3.70	3.70
190	3.80	3.80	3.80	3.80	3.80
195	3.90	3.90	3.90	3.90	3.90
200	4.00	4.00	4.00	4.00	4.00
205	4.10	4.10	4.10	4.10	4.10
210	4.20	4.20	4.20	4.20	4.20
215	4.30	4.30	4.30	4.30	4.30
220	4.40	4.40	4.40	4.40	4.40
225	4.50	4.50	4.50	4.50	4.50
230	4.60	4.60	4.60	4.60	4.60
235	4.70	4.70	4.70	4.70	4.70
240	4.80	4.80	4.80	4.80	4.80
245	4.90	4.90	4.90	4.90	4.90
250	5.00	5.00	5.00	5.00	5.00
255	5.10	5.10	5.10	5.10	5.10
260	5.20	5.20	5.20	5.20	5.20
265	5.30	5.30	5.30	5.30	5.30
270	5.40	5.40	5.40	5.40	5.40
275	5.50	5.50	5.50	5.50	5.50
280	5.60	5.60	5.60	5.60	5.60
285	5.70	5.70	5.70	5.70	5.70
290	5.80	5.80	5.80	5.80	5.80
295	5.90	5.90	5.90	5.90	5.90
300	6.00	6.00	6.00	6.00	6.00
305	6.10	6.10	6.10	6.10	6.10
310	6.20	6.20	6.20	6.20	6.20
315	6.30	6.30	6.30	6.30	6.30
320	6.40	6.40	6.40	6.40	6.40
325	6.50	6.50	6.50	6.50	6.50
330	6.60	6.60	6.60	6.60	6.60
335	6.70	6.70	6.70	6.70	6.70
340	6.80	6.80	6.80	6.80	6.80
345	6.90	6.90	6.90	6.90	6.90
350	7.00	7.00	7.00	7.00	7.00
355	7.10	7.10	7.10	7.10	7.10
360	7.20	7.20	7.20	7.20	7.20
365	7.30	7.30	7.30	7.30	7.30
370	7.40	7.40	7.40	7.40	7.40
375	7.50	7.50	7.50	7.50	7.50
380	7.60	7.60	7.60	7.60	7.60
385	7.70	7.70	7.70	7.70	7.70
390	7.80	7.80	7.80	7.80	7.80
395	7.90	7.90	7.90	7.90	7.90
400	8.00	8.00	8.00	8.00	8.00
405	8.10	8.10	8.10	8.10	8.10
410	8.20	8.20	8.20	8.20	8.20
415	8.30	8.30	8.30	8.30	8.30
420	8.40	8.40	8.40	8.40	8.40
425	8.50	8.50	8.50	8.50	8.50
430	8.60	8.60	8.60	8.60	8.60
435	8.70	8.70	8.70	8.70	8.70
440	8.80	8.80	8.80	8.80	8.80
445	8.90	8.90	8.90	8.90	8.90
450	9.00	9.00	9.00	9.00	9.00
455	9.10	9.10	9.10	9.10	9.10
460	9.20	9.20	9.20	9.20	9.20
465	9.30	9.30	9.30	9.30	9.30
470	9.40	9.40	9.40	9.40	9.40
475	9.50	9.50	9.50	9.50	9.50
480	9.60	9.60	9.60	9.60	9.60
485	9.70	9.70	9.70	9.70	9.70
490	9.80	9.80	9.80	9.80	9.80
495	9.90	9.90	9.90	9.90	9.90
500	10.00	10.00	10.00	10.00	10.00



Experiment 3: Measuring Surface Area

On 4/27, we confirmed this again, this time, measuring surface area too, found that as post heating time increases, the surface area decreases or stays relatively constant, while the height decreases.

Gemini suggested that this difference in results from the past 2 experiments, is due to part of the film coated into the well. When it is heated, the extra slack is released due to pressure and now the height is lower due to the slack.

	left max defl (nm)	right max defl (nm)	Left diameter	Left area	Left ylen	Left y height	Left surface area (from pyrd) (micrometers squared)	Right surface area (from pyrd) (micrometers squared)	Left surface area (corrected) (micrometers squared)	Right surface area (corrected) (micrometers squared)
0	243	237	8.74	5.09	6.13	9.81	75.49		59.34	-16.14999534
1	185	180	8.65	4.83	6.05	9.89	75.42		59.27	-16.14999534
2	155	151	9.03	4.55	5.51	9.89	75.39		59.34	-16.14999534
3	141	136	8.96	4.56	7.09	9.58	75.36		59.23	-16.14999534
4	150	140	8.70	4.72	6.12	9.79	75.42		59.27	-16.14999534
5	146	140	8.61	5.13	5.06	9.79	75.41		59.26	-16.14999534
7	130	124	8.66	4.8	4.66	9.79	75.37	80.01	59.22	63.86000466
9	121	113	8.78	4.65	5.35	9.75	75.35		59.20	-16.14999534
11	115	108	8.75	4.83	6.02	9.61	75.35		59.20	-16.14999534
13	119	110	8.69	5.11	4.72	9.6	75.36		59.21	-16.14999534
16	110	97	8.75	3.97	4.42	9.66	75.36		59.21	-16.14999534
19	102	91	8.69	4.46	6.12	9.58	75.35		59.20	-16.14999534
22	99	93	8.69	4.71	4.74	9.66	75.34		59.19	-16.14999534
26	91	80	8.66	5.2	4.55	9.58	75.34		59.19	-16.14999534
30	88	79	8.75	4.34	6.06	9.7	75.33		59.18	-16.14999534
34	84	74	8.72	5.02	3.52	9.61	75.3355		59.19	-16.14999534
39	75	68	8.69	5.15	3.76	9.62	75.33		59.18	-16.14999534

